Asymptotes

When we deal limit and that have a value of infinity or negative infinity, then we'll also take a brief look at vertical asymptotes. And also when we discuss about on polynomials and rational expression in the limit, we will also take a ~~look~~ horizontal asymptotes.

~~Asymptotes~~

Asymptotes

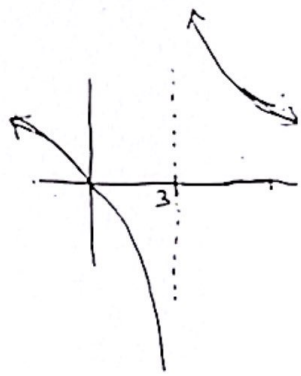
- Vertical Asymptote
- Horizontal Asymptote
- Oblique Asymptote

Vertical Asymptote

The function $f(x)$ will have a vertical asymptote at $x = a$ if we have any of the following limits at $x = a$

$$\lim_{x \to a^-} f(x) = \pm\infty, \quad \lim_{x \to a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a} f(x) = \pm\infty .$$

Example: $\lim_{x \to 3} \frac{2x}{x-3}$, $x - 3 = 0$ $\quad x = 3$



Since $\lim_{x \to 3^-} \frac{2x}{x-3} = -\infty$, $\lim_{x \to 3^+} \frac{2x}{x-3} = +\infty$

$\therefore$ ~~and~~ $\lim_{x \to 3} \frac{2x}{x-3}$ doesn't exist.

Which means $x = 3$ is the vertical asymptote of the function $f(x) = \frac{2x}{x-3}$.

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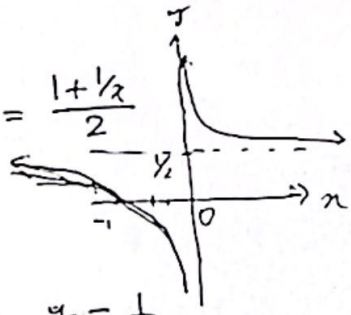
Horizontal Asymptote

The function $f(x)$ will have a horizontal asymptote at $y = L$ if either of the following are true.

$$\lim_{x \to \infty} f(x) = L \qquad \lim_{x \to -\infty} f(x) = L$$

Example: $f(x) = \frac{(x+1)}{2x} = \frac{x(1+1/x)}{2x} = \frac{1+1/x}{2}$



$$\lim_{x \to \infty} \frac{1+1/x}{2} = \frac{1}{2}$$

which means horizontal asymptote is $y = \frac{1}{2}$.

NOTE: If the degree of the numerator is greater than that of the denominator, then there is no horizontal asymptote.

e.g. $f(x) = \frac{x^2+3}{x+1}$, no horizonal asymptote.

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Problem: Let we defined a function $f(x) = \frac{x-1}{\sqrt{x^2}}$ and $\lim_{x \to +\infty} f(x) = a$ and $\lim_{x \to -\infty} f(x) = b$. Find the value of $(a-b)$.

Sol^n: $a = \lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} \frac{x-1}{\sqrt{x^2}} = \lim_{x \to +\infty} \frac{x-1}{|x|} = \lim_{x \to +\infty} \left[\frac{x}{|x|} - \frac{1}{|x|}\right]$

$= \lim_{x \to +\infty} \left[1 - \frac{1}{x}\right]$

$= 1$

since $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

$b = \lim_{x \to -\infty} f(x) = \lim_{x \to -\infty} \left[\frac{x}{|x|} - \frac{1}{|x|}\right]$

$= \lim_{x \to -\infty} \left[\frac{x}{-x} - \frac{1}{x}\right] = -1$

$= \cancel{\lim_{x \to -\infty}} \cancel{-1-0}$

$\therefore \quad a - b = 1 - (-1) = 2$ ※

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